

Arrays - Part 2

ForEach() →

→ executes a function for each elements.

```

let numbers = [10, 20, 30]
  numbers.forEach (function (num) {
    console.log(num)
  })
  
```

⇓

```

numbers.forEach! (num, idx) => {
  console.log(num, idx)
}
  
```

map(), reduce(), filter() - must

map()

↳ creates a new array by transforming

↳ creates a new array of every elements.

```
let numbers = [1, 2, 3, 4]
```

```
let dnumbers = numbers.map((ele) => {  
  return ele * 2
```

3)

```
console.log(dnumbers) → [2, 4, 6, 8]
```

± reduce() -

↳ reduces an array to a single value.

```
let numbers = [1, 2, 3, 4] →  
⑩ let sum = numbers.reduce((total, ele) => {  
  ⑩  
  return total + num  
  6 - 3 +  
  3, 6 }  
);
```

total = accumulator

ele → current element

0 → initial value

reduce ((total, ele) => { -3, 0 })

1) filter() → creates a new array containing elements that satisfy a condition =

Ex let numbers = [1, 2, 3, 4, 5, 9, 12, 16]

↓
Even no


```
let even = Numbers.filter( (ele) => {
    return ele % 2 === 0
})
```

Conso do - log (even) $\rightarrow$ 

find() → returns first element that
satisfy a condition.

no condition satisfy return $\rightarrow$ undefined

no condition return $\rightarrow$ undefined
satisfy

let numbers = [10, 20, 30, 40]

let result = numbers.find((ele) $\Rightarrow$ {
return num > 20

3) -

$\rightarrow$ (30)

findIndex() - index of first
matching element

$\rightarrow$ if no match return (-1)

numbers.findIndex((ele) $\Rightarrow$ ele > 20)
 $\downarrow$ $\downarrow$
index

some() $\rightarrow$ checks whether at least
one elements satisfy a condition
 $\rightarrow$ return $\rightarrow$ true | false

#1 Every() → Checks whether all elements satisfy a condition

→ true | false

let numbers = [10, 20, 30, 15, 10]

numbers.every((ele) => (ele >= 10))

↓
true

↓
all element match
↓
True

#1 Sort() → original array modify.
→ sorts an array —

→ for numbers, use a compare function

let numbers = [10, 11, 23, 43]

numbers.sort()

↓
ascending order

numbers.sort((a, b) => (a - b))

↓
ascending order

